UNIVERSITY OF TEXAS ARLINGTON
COLLEGE OF SCIENCE

Advanced Classical Mechanics

Lecture 11: Coupled Oscillations

Readings: Morin - Chapter 4, Marion Chapter 12

Amir Shahmoradi

Joshua Alexander Osborne

Department of Physics / College of Science

The University of Texas at Arlington

Arlington, Texas

Coupled Oscillations

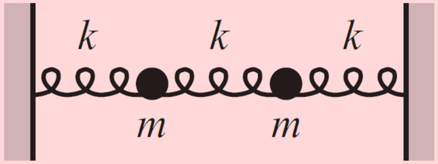
In the previous lectures, we examined the motion of an oscillator subjected to an external driving force. The discussion was limited to the case in which the driving force is periodic; that is, the driver is itself a harmonic oscillator. We considered the driver's action on the oscillator, but **we did not include the feedback effect of the oscillator on the driver**. In many instances, ignoring this effect is unimportant, but if two (or many) oscillators are connected in such a way that energy can be transferred back and forth between (or among) them, the situation becomes the more complicated case of coupled oscillations. Motion of this type can be quite complex (the motion may not even be periodic), but we can always describe the motion of any oscillatory system in terms of **normal coordinates**, which have the property that **each oscillates with a single, well-defined frequency**; that is, the **normal coordinates are constructed in such a way that no coupling occurs among them**, even though there is coupling among the ordinary (rectangular) coordinates describing the positions of particles. Initial conditions can always be prescribed for the system so that in the subsequent motion, only **one normal coordinate varies with time**. In this circumstance, we say that **one of the normal modes of the system has been excited**. If the system has n-degrees of freedom (e.g., n-coupled one-dimensional oscillators or $\frac{n}{3}$-coupled three-dimensional oscillators), there are, in general, n normal modes, **some of which may be identical**. The general motion of the system is a complicated superposition of all the normal modes of oscillation, **but we can always find initial conditions such that any given one of the normal modes is independently excited**. Identifying each of a system's normal modes allows us to construct a revealing picture of the motion, even though the system's general motion is a complicated combination of all the normal modes.

Problem 1. (Two Coupled Differential Equations) So far, we have dealt with motion consisting of a single position variable $x(t)$ as a function of time. Consider a more intricate scenario with two positions as functions of time, $x(t)$ and $y(t)$, which are related by a pair of "coupled" differential equations.

$$2\ddot{x} + \omega^2(5x - 3y) = 0 \quad ,$$
$$2\ddot{y} + \omega^2(5y - 3x) = 0 \quad .$$

where $\omega^2 > 0$ holds.

Problem 2. (Two masses, three springs) Consider two masses m, connected to each other and to two walls by three springs as in the figure. The three springs have the same spring constant k.
Find the most general solution for the positions of the masses as functions of time.
What are the normal coordinates?
What are the normal modes?



Lagrangian Approach to Coupled Oscillations

We found in the above problem that the effect of coupling in a simple system with two degrees of freedom produced two characteristic frequencies and two modes of oscillation. We now turn our attention to the general problem of coupled oscillations. Let us consider a conservative system described in terms of a set of generalized coordinates q_k and the time t. If the system has n degrees of freedom, then $k = l, 2, \dots ,n$. We specify that a stable equilibrium configuration exists for the system and that at equilibrium, the generalized coordinates have values q_{k0}. In such an equilibrium configuration, Lagrange's equations are satisfied by,

$$q_k = q_{k0}, \qquad \dot{q}_k = 0, \qquad \ddot{q}_k = 0, \qquad k = 1, 2, \dots, n$$

The Lagrangian at equilibrium therefore requires,

$$\left.\frac{\partial L}{\partial q_k}\right|_0 = \left.\frac{\partial T}{\partial q_k}\right|_0 - \left.\frac{\partial U}{\partial q_k}\right|_0 = 0$$

where the subscript 0 designates that the quantity is evaluated at equilibrium.
Assuming that the generalized coordinates and Cartesian coordinates mapping do not involve time explicitly, we can write the generalized kinetic energy as,

$$T = \frac{1}{2}\sum_{j,k} m_{jk}\dot{q}_j\dot{q}_k$$

Lagrangian Approach to Coupled Oscillations

The expansion of the potential energy in a Taylor series about the equilibrium configuration yields,

$$U(q_1, q_2, \ldots, q_n) = U_0 + \sum_k \left.\frac{\partial U}{\partial q_k}\right|_0 q_k + \frac{1}{2}\sum_{j,k} \left.\frac{\partial^2 U}{\partial q_j \partial q_k}\right|_0 q_j q_k + \cdots$$

The second term in the expansion must vanish to have an equilibrium at q_{k0}. Without loss of generality, we may choose to measure U in such a way that $U_0 = 0$. Then, if we **restrict the motion of the generalized coordinates to be small**, we may neglect all terms in the expansion containing products of the q_k of degree higher than second. This is equivalent to restricting our attention to **simple harmonic oscillations**, in which case only terms quadratic in the coordinates appear. Thus,

$$U = \frac{1}{2}\sum_{j,k} A_{jk} q_j q_k$$

where,

$$A_{jk} \equiv \left.\frac{\partial^2 U}{\partial q_j \partial q_k}\right|_0$$

Because the order of differentiation is immaterial (that is, U has continuous second partial derivatives), the quantity A_{jk} is symmetrical; that is, $A_{jk} = A_{kj}$. We have specified that the motion of the system is to take place in the vicinity of the equilibrium configuration, and we have shown that U must have a minimum value when the system is in this configuration.

Because we have chosen $U = 0$ at equilibrium, we must have, in general, $U > 0$. It should be clear that we must also have $T > 0$. We could consider the factors m_{jk} and A_{jk} in the above equations for the kinetic and potential energies respectively as the generalized mass and spring constants.

Lagrangian Approach to Coupled Oscillations

The equations of motion of the system with kinetic and potential energies at hand can be obtained from Lagrange's equation,

$$\frac{\partial L}{\partial q_k} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_k} = 0$$

But because T is a function only of the generalized velocities and U is a function only of the generalized coordinates, Lagrange's equation for the kth coordinate becomes,

$$\frac{\partial U}{\partial q_k} + \frac{d}{dt}\frac{\partial T}{\partial \dot{q}_k} = 0$$

From the kinetic and potential energy equations before, we can compute the Lagrangian derivatives as,

$$\left.\begin{aligned}\frac{\partial U}{\partial q_k} &= \sum_j A_{jk} q_j \\ \frac{\partial T}{\partial \dot{q}_k} &= \sum_j m_{jk} \dot{q}_j\end{aligned}\right\}$$

which yield the following equations of motion,

$$\sum_j (A_{jk} q_j + m_{jk} \ddot{q}_j) = 0$$

This is a set of n second-order linear homogeneous differential equations with constant coefficients. Because we are dealing with an oscillatory system, we expect a solution of the form,

$$q_j(t) = a_j e^{i(\omega t - \delta)}$$

Lagrangian Approach to Coupled Oscillations

where the a_j are real amplitudes and where the phase δ has been included to give the two arbitrary constants (a_j and δ) required by the second-order nature of each of the differential equations (only the real part of the right-hand side is to be considered). We accept without proof here that ω must also be real. These facts lead to the observation that the motion is indeed oscillatory and the equations of motion can be simplified to the form,

$$\sum_j (A_{jk} - \omega^2 m_{jk})\, a_j = 0$$

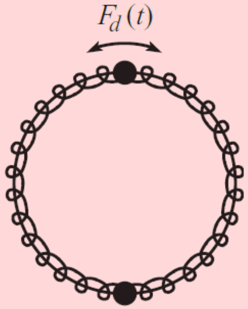
The above equation represents a set of n linear, homogeneous, algebraic equations that the a_j must satisfy. For a nontrivial solution to exist, the determinant of the coefficients must vanish:

$$\begin{vmatrix} A_{11} - \omega^2 m_{11} & A_{12} - \omega^2 m_{12} & A_{13} - \omega^2 m_{13} \cdots \\ A_{12} - \omega^2 m_{12} & A_{22} - \omega^2 m_{22} & A_{23} - \omega^2 m_{23} \cdots \\ A_{13} - \omega^2 m_{13} & A_{23} - \omega^2 m_{23} & A_{33} - \omega^2 m_{33} \cdots \\ \vdots & \vdots & \vdots \end{vmatrix} = 0$$

The equation represented by this determinant is called the **characteristic equation** or **secular equation** of the system and is an equation of degree n in ω^2 with generally n roots $\omega_r, r = 1, \ldots, n$ which are called the **characteristic frequencies** or **eigenfrequencies** of the system. In some situations, two or more of the ω_r can be equal; this is the phenomenon of **degeneracy**, to be further discussed in class.

Problem 3. (Two masses, three springs) Consider again the same problem as in Problem 2. Find the characteristic frequencies of the system using the Lagrangian Formalism discussed above.

Problem 4. (Driven mass on a circle)
Two identical masses m are constrained to move on a horizontal hoop.
Two identical springs with spring constant k connect the masses and wrap around the hoop. One mass is subject to a driving force $F_d \cos(\omega_d t)$.
Find the particular solution for the motion of the masses.



Problem 5. (Springs all over)
A mass m is attached to two springs with zero relaxed lengths.
The other ends of the springs are fixed at two points.
The two spring constants are equal.
The mass sits at its equilibrium position and is then given a
kick in an arbitrary direction.
Describe the resulting motion. (Ignore gravity, although you actually don't need to.)

Problem 6. (Star Springs)
A mass m is attached to n springs that have relaxed lengths of
zero. The other ends of the springs are fixed at various points in
space. The spring constants are $k_1, k_2, \ldots, k_n$. The
mass sits at its equilibrium position and is then given a kick in an
arbitrary direction. Describe the resulting motion.
(Ignore gravity, although you actually don't need to.)